

The Clock Stability Theorem: the Sub-Horizon Sound Speed of a Cuscuton-Clock MOND Sector Is Set by the Clock Rate

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Abstract

A relativistic MOND candidate built from a cuscuton clock τ , a dynamical scalar χ with a projected-gradient (MOND) sector $sW(Y, \tau)$, and a cubic coupling $\gamma X \square \chi$ has a homogeneous cosmological branch on which the clock sector gravitates like dust. We derive, in the uniform-clock gauge and the sub-horizon limit, the sound speed of the clock-scalar mode: $c_s^2 = [2P_X(1-D) - 2s_0W_Y]/[B(1-D)]$ with $B = 2P_X + 4q^2P_{XX}$ and $D = 2q^2W_Y/W$. The cuscuton constraint and the lapse response enhance the MOND gradient term by $1/(1-D)$; the naive k -essence form is wrong by factors of -28 to $+4$. Under the candidate's coefficient closure the result collapses to $c_s^2 = (1-s_0)m_{\text{rel}}/(2-m_{\text{rel}})$, where $s_0 = \dot{\tau}$ is the clock rate relative to proper time and m_{rel} the logarithm margin of the closure. Verified against the candidate's exact 6×6 finite-wavelength transfer operator at ten points of its branch to 1.1×10^{-3} (general form) and 2.2×10^{-3} (closure form), the residual being the $\gamma = 10^{-6}$ terms dropped in the derivation. Consequences, each machine-checked in Lean 4: the sector is gradient-stable if and only if the clock does not run faster than proper time ($s_0 \leq 1$; on the branch s_0 crosses 1 once, at $a = 0.29$); a sound speed small enough for the Lyman- α forest ($c_s^2 \leq 10^{-9}$) forces $m_{\text{rel}} \leq 2 \times 10^{-9}/(1-s_0) \sim 10^{-8}$, five orders below the branch; and no sound speed can supply the host-mass-dependent dark-fraction depletion that galaxies require, since the forest-compatible $c_s = 9.5 \text{ km s}^{-1}$ has a Jeans length below 7 kpc at galactic densities. The theorem turns the candidate's stability into an a-priori condition on its coefficient history and shows that its depletion mechanism, if any, must be dynamical rather than acoustic.

1 Setting and attribution

The action, its coefficient closure, its sourced homogeneous branch and its finite-wavelength linear transfer operator are the work of the collaborating field-theory programme in the repository [1] (directory `qwen_claude_field_theory/closure_2026/clock_response_repair_2026/`, commit `c5ffc6b59`), whose status was published in [2]. Nothing of that programme is modified here: its modules are imported read-only and its archived branch is used as data. This note contributes the analytic derivation of the sub-horizon sound speed, its numerical verification against that operator, the Lean 4 certificates, and the consequences.

In signature $(-+++)$ with $c = 1$ and $M^2 > 0$, the frozen action is

$$S = \int d^4x \sqrt{-g} \left[\frac{M^2}{2} (R - 2\Lambda) + P(X, \tau) - V(\tau) + sW(Y, \tau) + \gamma X \square \chi \right] + S_r + S_b, \quad (1)$$

$$s = \sqrt{-\nabla_\mu \tau \nabla^\mu \tau}, \quad n_\mu = -\partial_\mu \tau / s, \quad Q = n^\mu \partial_\mu \chi, \quad Y = (g^{\mu\nu} + n^\mu n^\nu) \partial_\mu \chi \partial_\nu \chi, \quad X = Q^2 - Y, \quad (2)$$

with a perfect-fluid radiation surrogate S_r and irrotational baryon dust S_b minimally coupled. The action is linear in s : the clock is a cusciton [3] and its field equation is a constraint. The coefficient closure evaluates, on the branch, $W = U$, $W_Y = d$, $P_X = Ud/m$, $P_{XX} = 2Ud^2/m^2$ with $m = U - 2dq^2$ the “logarithm margin” of $P = -\frac{U}{2} \log \frac{U-2dX}{U-2d\bar{q}^2} + \dots$, $q = \dot{\chi}$, $m_{\text{rel}} = m/U$; the functions U, d are fixed histories, reconstructed rather than derived, which is the candidate’s stated limitation.

2 The linear scalar sector

In the uniform-clock gauge $\delta\tau = 0$ with shear-free spatial gauge, the scalar perturbations are the lapse α , shift b , volume z , the scalar $\sigma = \delta\chi$ and the matter variables; Fourier amplitudes at wavenumber k , $\mathcal{L} = -k^2/a^2$, $\delta K = 3\dot{z} - 3H\alpha - kb$, $v = \delta Q = \dot{\sigma} - q\alpha$. The clock constraint and the scalar equation of the candidate’s derivation are

$$\delta E_\tau = 2qP_{X\tau}v - W\delta K + 2qW_Y\mathcal{L}\sigma = 0, \quad (3)$$

$$\delta D j = \dot{\delta j} + 3H\delta j + j\delta K - \alpha\dot{j} + [-2P_X + 2s_0W_Y + 2\gamma(\dot{q} + 3Hq)]\mathcal{L}\sigma - 2\gamma q\mathcal{L}v = 0, \quad (4)$$

with $\delta j = (B - 12\gamma Hq)v - 2\gamma q^2\delta K + 2\gamma q\mathcal{L}\sigma$, $B = 2P_X + 4q^2P_{XX}$, $j = 2qP_X - 6\gamma Hq^2$, and $s_0 = \dot{\tau}$ the background clock rate, solved rather than set to one.

3 Derivation

Take $k \gg aH$, keep the leading powers of k , and set $\gamma \rightarrow 0$ in the derivation (it is kept in the operator used for verification). Four steps.

(i) *The cusciton constraint fixes the slicing at $O(k^2)$.* From (3),

$$\delta K = \frac{2qP_{X\tau}v + 2qW_Y\mathcal{L}\sigma}{W} \equiv Fv + G\sigma, \quad G = -\frac{2qW_Y}{W} \frac{k^2}{a^2}. \quad (5)$$

(ii) *The momentum constraint makes the shift $O(k)$.* $kb = -\delta K - 3\mathcal{M}/(2M^2)$ with $\mathcal{M}$ the $O(1)$ momentum density, so $b = O(k)\sigma$ and the gauge-invariant potentials remain small: $\Psi = -z - HS$ with $S = -a^2b/k$ cancels at leading order.

(iii) *The lapse responds at $O(1)$.* The lapse equation has the form $(qG + p + \dots)\alpha = -[(\dot{G} + 2HG)\sigma + Gv] + O(1)$ with $p = k^2/a^2$; since $G \propto p$, α is $O(1)$:

$$\alpha = \frac{Dv + (\dot{D} - \frac{\dot{q}}{q}D)\sigma}{q(1-D)}, \quad D \equiv \frac{2q^2W_Y}{W}, \quad (6)$$

so that $v = \dot{\sigma} - q\alpha$ gives $v = (1-D)\dot{\sigma} + O(\sigma)$: the constraint and the lapse renormalise the kinetic coefficient of σ by $(1-D)$.

(iv) *The scalar equation.* At $O(k^2)$, (4) contains $[2P_X - 2s_0W_Y](k^2/a^2)\sigma$ from the explicit gradient terms and $jG\sigma = -(2qW_Yj/W)(k^2/a^2)\sigma$ from the constraint, with $j = 2qP_X$; the kinetic term is $B\dot{v} = B(1-D)\ddot{\sigma} + \dots$. Hence

$$c_s^2 = \frac{2P_X(1-D) - 2s_0W_Y}{B(1-D)}, \quad B = 2P_X + 4q^2P_{XX}, \quad D = \frac{2q^2W_Y}{W} \quad (7)$$

The MOND gradient term $s_0 W_Y$ is enhanced by $1/(1 - D)$ relative to the naive k -essence form $(2P_X - 2s_0 W_Y)/B$.

Closure. With $W = U$, $W_Y = d$, $P_X = Ud/m$, $P_{XX} = 2Ud^2/m^2$ and $m = U - 2dq^2$: $1 - D = m/U = m_{\text{rel}}$, $2P_X(1 - D) = 2d$, and $B(1 - D) = 2d(2U - m)/m$, so

$$c_s^2 = (1 - s_0) \frac{m_{\text{rel}}}{2 - m_{\text{rel}}} \quad (8)$$

4 Verification against the exact operator

The candidate's 6×6 first-order transfer operator (state $\sigma, v, r_1, v_r, \theta_1, \delta\rho_b$; its reconstructed fields satisfy the original eight Euler equations to 10^{-8} in the candidate's own tests) was evaluated at ten archived samples of its backward, radiation-majority branch and its eigenvalues taken at $k = 300, 3000, 30000 \gg aH$; a propagating pair has $\lambda^2 = -c_s^2 k^2/a^2$. The perfect-fluid radiation pair returns $c_s^2 = 1/3$ to four digits at every sample and the clock pair is k -converged to 0.3% (script L185 of [1]). Table 1 compares the operator value with (7) and (8) (script L186). The

Table 1: Clock sound speed on the candidate's branch: exact operator versus the derived forms.

a	s_0	m_{rel}	operator ($k = 3 \times 10^4$)	eq. (7)	eq. (8)	dev. (7), (8)
1.000	1.032	9.34×10^{-2}	-1.554×10^{-3}	-1.552×10^{-3}	-1.557×10^{-3}	$-1.1 \times 10^{-3}, +2.2 \times 10^{-3}$
0.750	1.072	7.36×10^{-2}	-2.741×10^{-3}	-2.740×10^{-3}	-2.742×10^{-3}	$-2.3 \times 10^{-4}, +6.7 \times 10^{-4}$
0.563	1.173	5.60×10^{-2}	-4.989×10^{-3}	-4.988×10^{-3}	-4.990×10^{-3}	$-5.0 \times 10^{-5}, +2.8 \times 10^{-5}$
0.422	1.413	3.79×10^{-2}	-7.961×10^{-3}	-7.961×10^{-3}	-7.962×10^{-3}	$-1.2 \times 10^{-5}, +1.7 \times 10^{-5}$
0.317	1.118	2.02×10^{-2}	-1.198×10^{-3}	-1.198×10^{-3}	-1.199×10^{-3}	$-2.7 \times 10^{-5}, +5.5 \times 10^{-5}$
0.238	0.851	9.05×10^{-3}	$+6.780 \times 10^{-4}$	$+6.780 \times 10^{-4}$	$+6.777 \times 10^{-4}$	$+1.5 \times 10^{-5}, -4.4 \times 10^{-5}$
0.178	0.771	3.89×10^{-3}	$+4.454 \times 10^{-4}$	$+4.454 \times 10^{-4}$	$+4.453 \times 10^{-4}$	$+8.2 \times 10^{-6}, -3.6 \times 10^{-6}$
0.134	0.740	1.65×10^{-3}	$+2.145 \times 10^{-4}$	$+2.145 \times 10^{-4}$	$+2.144 \times 10^{-4}$	$+1.1 \times 10^{-5}, -4.2 \times 10^{-5}$
0.100	0.723	6.98×10^{-4}	$+9.662 \times 10^{-5}$	$+9.662 \times 10^{-5}$	$+9.657 \times 10^{-5}$	$+2.9 \times 10^{-5}, -5.1 \times 10^{-5}$

general form agrees to 1.1×10^{-3} at worst and the closure form to 2.2×10^{-3} ; the deviations scale with $\gamma/d \sim 10^{-3}$ and are the cubic-coupling terms dropped in the derivation. Two adversarial checks: the candidate's independent forward background object reproduces the $a = 1$ value to 2×10^{-4} , and the growing eigenvector at $a = 0.42$ has 99.8% of its norm in the (σ, v) components, so the unstable mode is the clock-scalar mode and not dust or radiation.

5 Theorems

The following are theorems of the repository's Lean 4 file (`Mondlean.lean`, 105 theorems, no `sorry`; each depends only on `propext`, `Classical.choice`, `Quot.sound`).

Theorem 1 (`clock_sound_speed_closure`). For reals U, d, q, m, s_0 with $U, d, m \neq 0$, $2U - m \neq 0$ and $m = U - 2dq^2$,

$$\frac{2\frac{Ud}{m}(1 - \frac{2q^2d}{U}) - 2s_0d}{(2\frac{Ud}{m} + 4q^2\frac{2Ud^2}{m^2})(1 - \frac{2q^2d}{U})} = (1 - s_0) \frac{m/U}{2 - m/U}.$$

Theorem 2 (`clock_gradient_stability_iff`). For $0 < m_{\text{rel}} < 2$: $0 \leq (1 - s_0) m_{\text{rel}} / (2 - m_{\text{rel}}) \iff s_0 \leq 1$.

Theorem 3 (`clock_softness_forces_margin`). For $0 < m_{\text{rel}} < 1$, $\delta > 0$, $s_0 \leq 1 - \delta$ and $(1 - s_0) m_{\text{rel}} / (2 - m_{\text{rel}}) \leq \varepsilon$: $m_{\text{rel}} \leq 2\varepsilon/\delta$.

The calculus of Section 3 (the $k \rightarrow \infty$ limit and the constraint elimination) is not formalised; the theorems certify the closure algebra and the two inequalities, and the numerical verification of Section 4 certifies that the algebra is the candidate's.

6 Consequences

Stability is a clock-rate condition. By Theorems 1–2 the sector is gradient-stable exactly when the clock does not run faster than proper time. On the candidate's branch s_0 is 1.03–1.41 for $a \geq 0.32$ (unstable, real eigenvalues $\pm |c_s|k/a$ with $|c_s|$ up to 0.089, unbounded growth in k) and 0.85–0.72 for $a \leq 0.24$ (stable); it crosses 1 exactly once, at $a = 0.29$. This is a consistency condition that can be imposed on the coefficient history a priori; it is not a fit to data.

The forest bound in closed form. On the repository's CMB/forest tooling (CLASS with cold matter replaced by a $w \simeq 0$ fluid of sound speed c_s^2 ; L185), the third acoustic peak is restored for $c_s^2 \leq 10^{-2}$, so the early, stable part of the branch ($c_s^2 = 10^{-4}$ – 7×10^{-4}) would cluster on acoustic scales; but the Lyman- α power at $k = 5 h \text{ Mpc}^{-1}$, $z = 3$ is within 10% of ΛCDM only for $c_s^2 \leq 10^{-9}$. By Theorem 3 with $\varepsilon = 10^{-9}$ and the branch's own s_0 , the margin must satisfy $m_{\text{rel}} \leq 7 \times 10^{-9}$ – 1.3×10^{-8} , against 7×10^{-4} – 9×10^{-3} on the branch: the background would have to sit within 10^{-8} of the logarithm singularity throughout $z \leq 3$.

No acoustic depletion. The repository's necessity certificate requires the dark fraction to fall with host mass ($f \leq 0.105$ inside spirals of $1.2 \times 10^{10} M_\odot$, ≥ 0.988 at recombination). A sound speed cannot supply this: the forest-compatible $c_s = \sqrt{10^{-9}} c = 9.5 \text{ km s}^{-1}$ has a linear Jeans length of 6.7, 2.1, 0.7 kpc at $\rho = 10^{-25}$, 10^{-24} , $10^{-23} \text{ g cm}^{-3}$, so a fluid soft enough for the forest clusters inside galaxies. The depletion must be dynamical; the dynamical routes computed to date (two-body decay or kicks) are excluded by a lifetime pincer, forest $\tau \geq 41 \text{ Gyr}$ against galaxies $\tau \leq 20 \text{ Gyr}$.

7 What this does not claim

No statement is made about the existence of a viable coefficient history, about the candidate's static (galactic) branch, about ghosts (Theorem 2 concerns the gradient sign only; $B > 0$ on the branch), or about the CMB beyond the fluid proxy. The derivation is a frozen-coefficient WKB limit without mode coupling and drops γ ; the branch is the candidate's dimensionless one, not calibrated to recombination, and only the dimensionless c_s^2 is carried across. The Jeans estimate is linear and order-of-magnitude.

8 Reproducibility

Scripts `fable_independent_2026/L185_astra_clock_sound_speed.py` and `L186_clock_stability_theorem.py` with their committed outputs (commits `2dbfc6013`,

cd1fc0b88); the Lean file `fable_independent_2026/lean_2026/Mondlean.lean` (lake env lean Mondlean.lean; #print axioms on the three names above). The candidate’s operator and branch: the collaborating programme’s directory named in Section 1 at commit c5ffc6b59.

References

- [1] The research repository, <https://github.com/carlzimmerman/zimmerman-formula> (2026).
- [2] C. P. Zimmerman, *A Primordial-Clock Relativistic MOND Candidate: Status Report*, doi:10.5281/zenodo.22699828 (2026).
- [3] N. Afshordi, D. J. H. Chung, G. Geshnizjani, *Cuscuton: A causal field theory with an infinite speed of sound*, Phys. Rev. D **75**, 083513 (2007).
- [4] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. **270**, 365 (1983).